Team Number:	apmcm2311106
Problem Chosen:	A

2024 APMCM summary sheet

With the rapid development of machine learning, the realization of target crop identification in facility agriculture through image recognition methods is no longer a mere dream.

For **Question 1**, we developed a **grid-based object detection model** using the **YOLOv5x**. We first manually annotated samples using **LabelImg** and performed data augmentation (Figure 4), increasing the sample size to 1200. Then we found that the model's precision and recall rates are excellent (Figure 6). Upon manual review, the final trained model can accurately identify apples (Figure 5), and a histogram of its distribution is shown in **Figure 7**.

For **Question 2**, based on the images identified by **Model 1**, we obtained the sample bounding box coordinates with the bottom left corner as the origin through coordinate transformation and calculated the sample's **center coordinates** (Figure 8). Finally, we plotted a scatter plot of the coordinate distribution as shown in **Figure 9**.

For **Question 3**, we established a **maturity recognition model**, using **hue and saturation** from the HSV model as the criteria for judging the maturity of sample apples. We first maximized the color information of the pre-processed data, then transformed the obtained hue and saturation sample frequencies into feature vectors, drew an **undirected weighted graph**, and finally clustered **three types of apple maturity characteristics**, as shown in the histogram distribution in **Figure 15**.

For **Question 4**, we developed a **multi-modal quality measurement model**. We use a combined model based on two-dimensional area and depth estimation of the actual area to estimate the weight of the apple. we first identified the actual depth in the sample images using the **ZoeDepth-M12-NK** model. Then, we calculated the volume of the apples, and finally using **scaling**, we calculated the mass of each apple, as shown in the histogram distribution in **Figure X**.

For **Question 5**, we found that the Model 1 established in Q1 still performed well. Thus, after standardizing the data, we established a sample set of 1000 images per variety. We then tested the trained model on samples from Attachment 3. The final model recognition effect was good, and the result is shown in Figure 10.

Keywords: YOLO Spectral Clustering Depth Estimation

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I. Introduction

1.1 Problem Background

As the world's largest producer of apples, China has reached a total apple production of 45.97 million tons in 2021 [1]. Apple harvesting, an essential part of the agricultural supply chain, has always faced challenges such as short seasons, concentrated periods, high labor intensity and other issues like high labor costs [1]. In recent years, the rural to urban migration and industrial transformation have led to a significant loss of rural labor, resulting in a severe labor shortage in the fruit and vegetable harvesting industry. This situation exacerbated the crisis in apple harvesting.

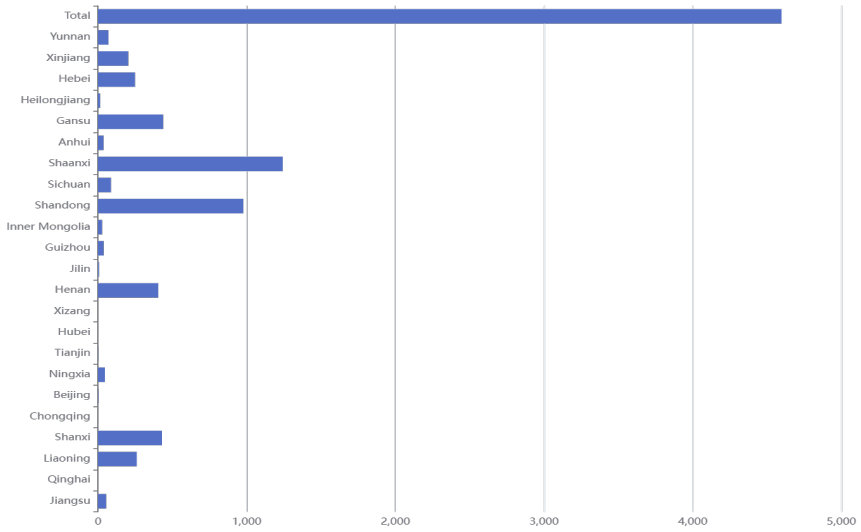


Figure 1 2021 Apple Production in China (Resource: http://www.moa.gov.cn/)

To tackle the challenges facing the industry, central government has continuously emphasized the importance of developing facility agriculture in the policy documents. With the advancement of artificial intelligence technology, automated identification and harvesting of apples using machines have become feasible. However, the challenge of enabling robots to recognize the number, location, and size of apples through image recognition still needs to be resolved.

1.2 Restatement of Problem

To establish a comprehensive model which could accurately identify and calculate some features of apples through the images, we need to complete the following questions:

• **Apple Counting:** Establish a mathematical model which could counts the number of apples in each image provided by *Attachment 1* and create a histogram to show distribution of all apples.

• **Apple Position Estimation:** Locate the apples in each image and plot a scatter graph showing the distribution of apple positions across *Attachment 1* by establishing a coordinate system.

• **Apple Maturity Estimation:** Establish a mathematical model which could calculate the maturities of apples in each image and draw a histogram of the maturity distribution of all apples.

• **Apple Quality Estimation:** Estimate the qualities of each apples provided by *Attachment 1*. And then we need to draw a histogram of quality distribution of all apples.

• **Apple Recognition:** Identify apples in *Attachment 3* and create a histogram to display the distribution of ID numbers for all apple images in that attachment.

1.3 Our work

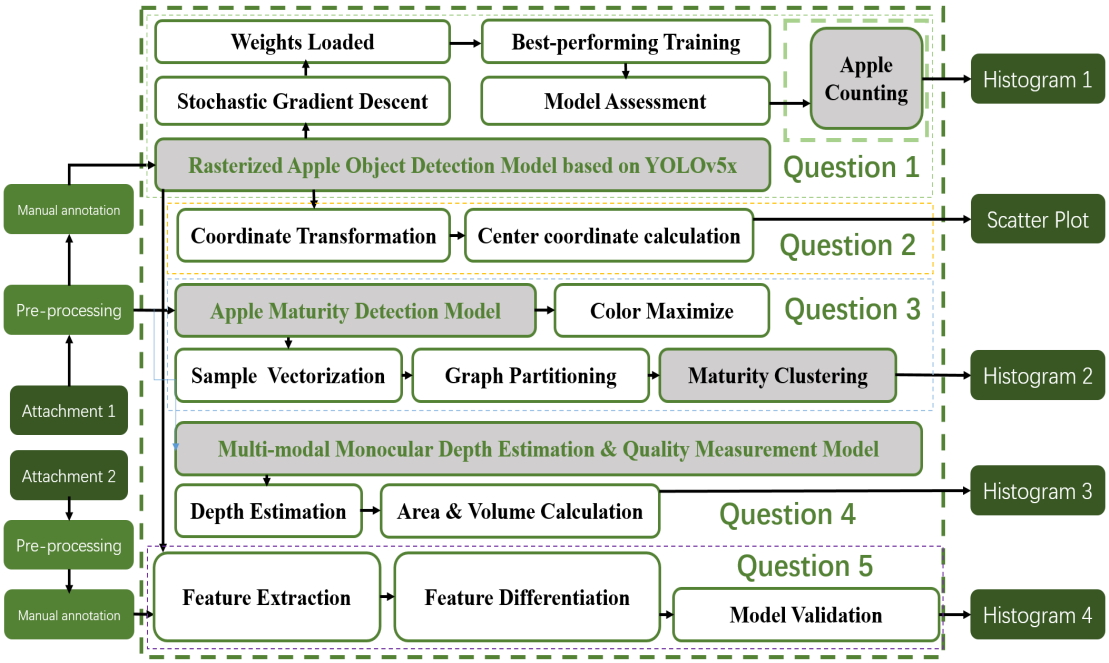


Figure 2 Our Work

II. Assumptions & Notations

2.1 Assumptions

To simplify our modeling, we make the following assumptions:

• **Assumption 1** When calculating the maturity features of apples, we assume the apples are perfectly circular. The shape of the apple does not significantly affect the determination of maturity, and unifying the shape can better assist us in segmenting the image samples.

• **Assumption 2** In calculating the actual area of an apple, we assume the apple's cross-section is either the inscribed circle or inscribed ellipse of the bounding box. This allows us to calculate the area of the apple with minimal error.

• **Assumption 3** We default apples with abnormal shape indices to a circular shape. This helps to eliminate the incomplete labeling of the bounding box that may occur due to leaf occlusion.

2.2 Notations

Symbol	Description
mAP	mean Average Precision
P	precision
R	recall rate
SI	Shape Index
SNN	Spiking Neural Network
$YOLO$	You Look Only Once
IoU	Intersection over Union

Table 1 Key notations used in this paper

III. Model 1-Rasterized Apple Object Detection Model based on YOLOv5x

Compared to other object detection models that use Region Proposal to aid in target localization and recognition (such as R-CNN and SPP Net), YOLO model can input the entire image into the initial layer and directly regress the target bounding boxes of objects, achieving the effect that is called ”one-stage”. We ultimately selected **the YOLOv5x model** for identifying and locating apples. Because compared to other algorithms, it features fast speed, high accuracy, high flexibility and ease of use and deployment.

3.1 Description of the Model

The YOLO (You Only Look Once) model is a state-of-the-art, real-time object detection system known for its speed and accuracy. It frames object detection as a single regression problem, directly predicting bounding boxes and class probabilities from full images in one evaluation.

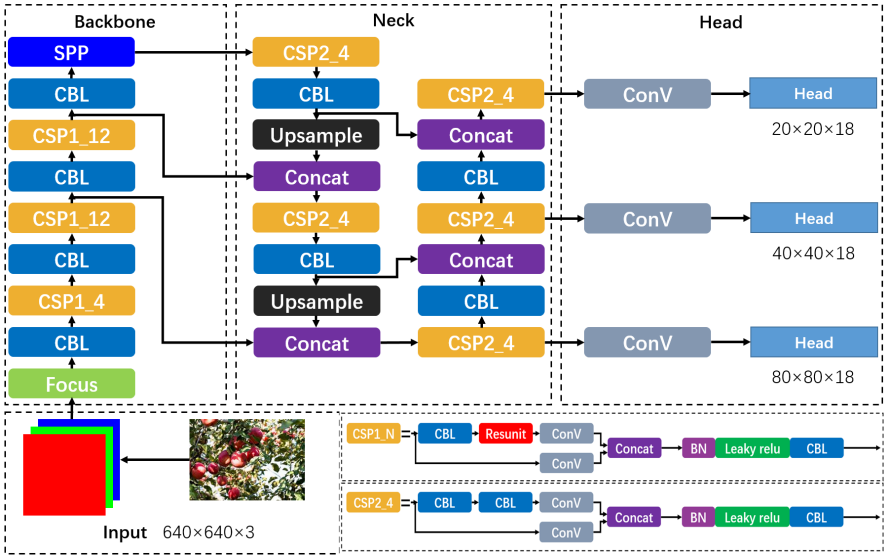


Figure 3 YOLOv5x network structure

The YOLOv5 model is composed of four main parts: the input end, Backbone, Neck and Head. Its fundamental components contain Focus, CBL, CSP1_N, CSP2_N and SPP. The CBL is formed by a combination of Convolution (ConV), Batch Normalization (BN), and Leaky ReLU. Compared to its predecessors, YOLOv5 introduces Mosaic data augmentation at the input stage, employs the Focus structure for image slic-

ing, and divides the CSP structure into 1_N and 2_N, which are applied in the Backbone and Neck. These enhancements enrich the dataset and improve the model's robustness. YOLOv5x, an iteration of YOLOv5, alters the 'depth_multiple' and 'width_multiple' parameters, resulting in higher training accuracy and effectively addressing the issue of inadequate precision inherent in the YOLO model. The specific comparison is as shown in Table 2.

Model	size	mAP^{val} 0.5:0.95	Speed(ms) GPU b1	params(M)
YOLOv5n	640	28.0	45	1.9
YOLOv5s	640	37.4	98	7.2
YOLOv5m	640	45.4	224	21.2
YOLOv5l	640	49.0	430	46.5
YOLOv5x	640	50.7	766	86.7

Table 2 The comparison of YOLOv5 models

The loss function of YOLOv5x is composed of three parts: classification loss, object confidence loss, and bounding box regression loss. The code for each part and the total loss function are as follows:

$$L_{cls} = -\sum_{i} y_i \log(\hat{y}_i)$$

$$L_{obj} = -\sum (y \log(\hat{y}) + (1-y)\log(1-\hat{y}))$$

$$L_{iou} = 1 - \text{IoU} + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha \cdot v$$

$$L_{total} = \lambda_{cls} L_{cls} + \lambda_{obj} L_{obj} + \lambda_{iou} L_{iou}$$

L_{cls} is the classification loss function, where y is the true label of the target category, and $\hat{y}_i$ is the probability predicted by the model. L_{obj} is the object confidence loss function, where y is the true label for the presence of an object, and $\hat{y}$ is the confidence predicted by the model. L_{iou} is the bounding box loss function. IoU is the intersection over union of the predicted box and the actual box, ρ is the Euclidean distance between the center of the predicted box and the actual box, c is the diagonal

length of the minimum enclosing area containing both boxes, v is a term related to the aspect ratio, and α is its weight.

We chose the $CIoU$ function as the bounding box loss function for this model because, compared to other IoU functions, the $CIoU$ function takes into account the overlap of bounding boxes, center distance, and aspect ratio. It addresses the issue of non-overlapping situations and thus has better convergence performance.

3.2 Data Processing

Model 1 sources its data from *Attachment 1*, which, upon inspection, contains 200 photos from various websites. Some of these photos don't feature apples. And we also find that the diverse angles and lighting conditions of these photos could add complexity to apple recognition.

Subsequently, we processed these sample images, including resizing all image data to 640×640 pixels and filtering out disruptive images. Using **LabelImg** software, we annotated all apples in the 200 images of *Attachment 1*, storing the category and location information of the apples in corresponding XML files. To maintain the accuracy of model discrimination, we opted not to annotate apples that were too small or heavily obscured [2].

Given that deep learning algorithms require a substantial amount of sample data for optimal performance, and the limited quantity of fruit images in the attachment, we employed data augmentation to expand the dataset. Augmentation methods included translation, obfuscation, rotation, mirroring, cutting, brightness enhancement, brightness reducing, random covering, and combinations of these effects. After augmentation, the number of images is expanded to 1,200. Subsequently, we divided the dataset into training and testing sets in the ratio of 8:2. The effects of image augmentation processing are illustrated in Figure 4.

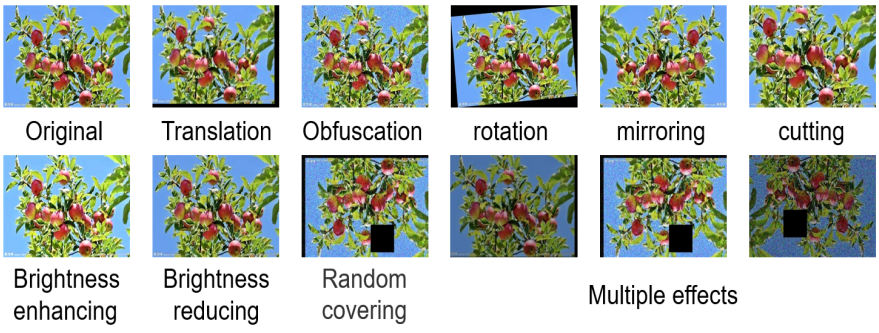


Figure 4 Augmentation Method

3.3 Experimental Environment

The experimental environment configuration for this paper is shown in the following table:

Name	Type
CPU	Intel Core I9 12900K
GPU	RTX 3080Ti
RAM	128G
SSD	SSD 4T
Motherboard	Z690

Table 3 The experimental environment configuration

3.4 Training & Experimental Results

3.4.1 *Training Process*

The training process utilized the default hyper-parameters of YOLOv5x, with initial images sized at 640×640 pixels. Weights were loaded using a pretrained model, and the network was optimized via Stochastic Gradient Descent to accelerate convergence, setting the learning rate at 0.01 and the weight decay factor at 0.005. The Batch-size was set to 8, and the process iterated for 40 rounds, selecting the best-performing model iteration for the *best.pt* training.

3.4.2 *Performance Evaluation*

This paper employs precision (P), recall rate (R) and F1-score as metrics to measure the precision of object detection. The formulas for these metrics are shown below, and the evaluation results are illustrated in Figure 6.

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

$$F1 = 2 \times \frac{P \cdot R}{P + R}$$

$$mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i$$

From the model evaluation metrics, it is evident that the model performs well in the target detection of apples. Confusion Matrix displays the model's actual selection cases, where the results show that the number of samples correctly predicted as "apple" reached 0.91 when the actual label was "apple". Upon manual inspection of the target annotation, we found that, despite some apples being obscured by leaves or having a greener color, the model was still able to accurately identify them to a certain extent. Similarly, the model's discrimination of negative cases was also significant. Thus, both from the metrics and practical observations, we could conclude the model's predictive performance is quite satisfactory.



Figure 5 Recognition of Non-standard Samples and Negative Cases

Similarly, based on the data shown in Figure 6, the model's detection performance appears strong. The P_curve exhibits a characteristic bending towards the upper-left corner, and the slope in the ROC curve is sufficiently steep. Additionally, the PR_curve closely approaches the upper-right corner. All three graphs collectively indicate good performance of the model.

3.4.3 *Question 1-Apple Counting*

Figure 7 vividly demonstrates the outstanding performance of the YOLOv5x model in recognizing apples. After manually verifying the model's effectiveness and making appropriate parameter adjustments, we have derived a histogram table regarding the distribution of apple counts. In this table, the horizontal axis represents the number of apples, while the vertical axis indicates the frequency of images with that specific count

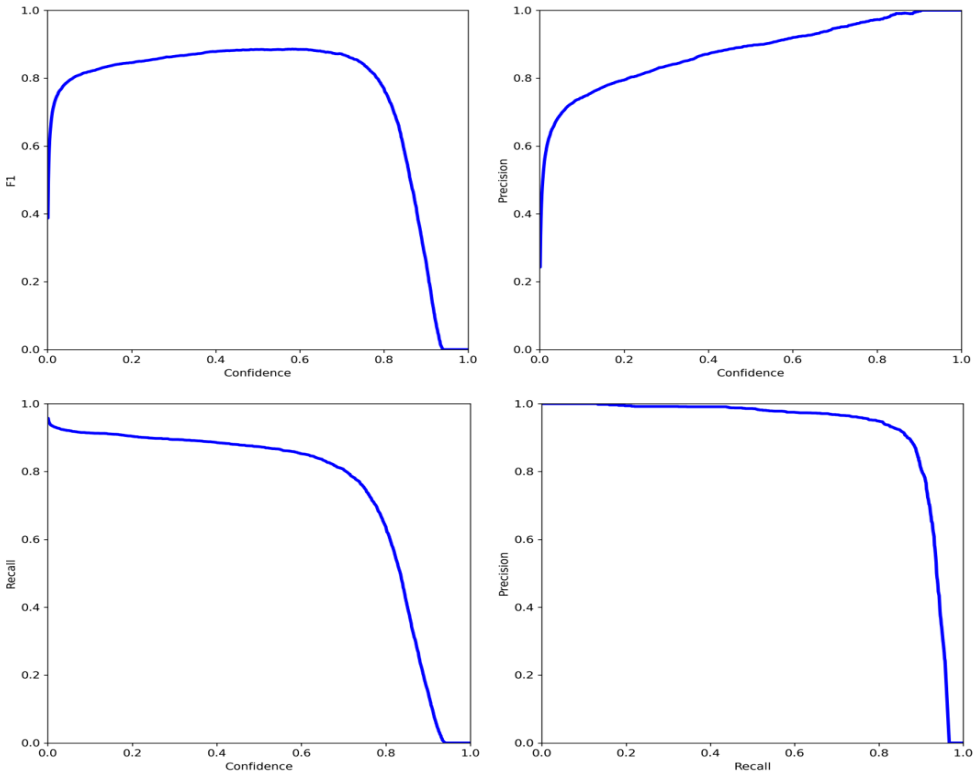


Figure 6 F1/P/R/PR curves

of apples. The results show that in most images, the number of apples ranges between **0-10**, and **almost all images feature apple counts within the 0-30 range**.

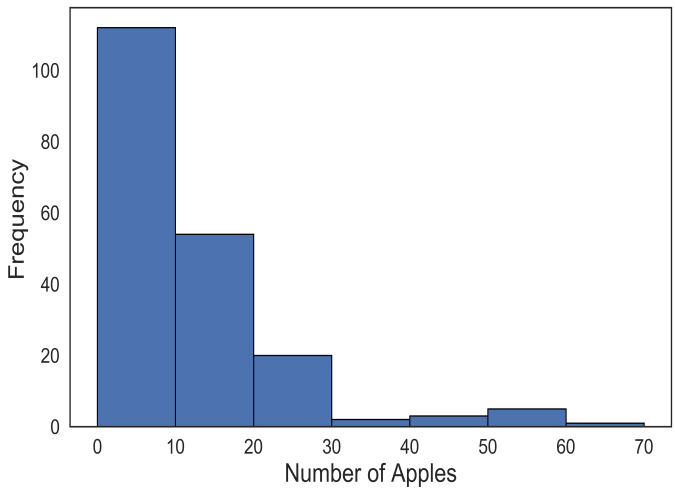


Figure 7 Histogram of Apple counts per Image

3.4.4 *Question 2-The position of Apples*

There are several common methods for representing bounding box coordinates in machine learning, such as Pascal VOC, COCO, and YOLO. For **Pascal VOC**, the coordinate encoding is $[x_{min}, y_{min}, x_{max}, y_{max}]$, representing the coordinates of the top-left and bottom-right corners of the bounding box. We locate apples using the Pascal VOC coordinate encoding. Considering that scatter plots require two-dimensional coordinates of apples with the origin at the bottom-left corner of the image, and since YOLO's default coordinates originate from the top-left corner, we performed a coordinate transformation, and resulted in x_{center} and y_{center} coordinates with the origin at the bottom-left corner of the image. The formulas are as follows:

$$y_{center} = h - \frac{\mathrm{y_{max}} + \mathrm{y_{min}}}{2}$$
$$x_{center} = \frac{\mathrm{x_{max}} - \mathrm{x_{min}}}{2} + x_{min}$$

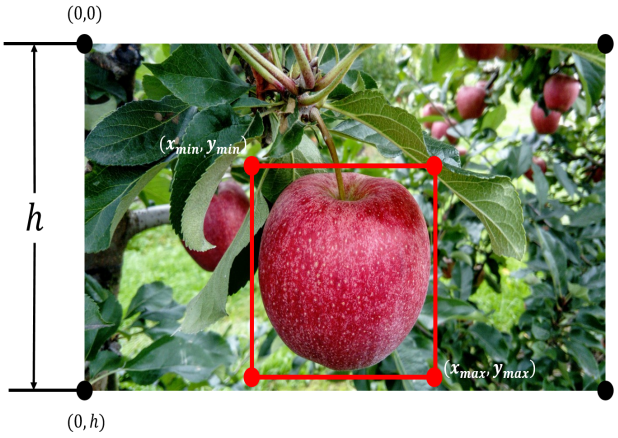


Figure 8 Pascal VOC coordinates

Finally, we obtained the center coordinates of all recognizable apples in the 200 images of *Attachment 1*. These coordinates are displayed in a scatter plot, as shown in Figure 9.

From the image, we can discern that the distribution of the apple centers is **fairly uniform**, with a general tendency to be concentrated towards the center of the image. The distribution of points along the edges is comparatively sparse.

3.4.5 *Question 5-Apple Recognition*

To solve **Question 5**, we need to identify all images containing apples in *Attachment 3* by extracting image features from *Attachment 2*. Given the large number of samples

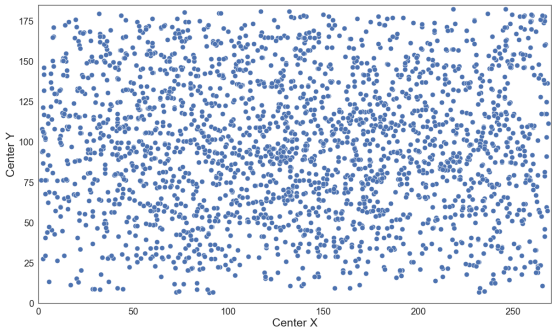


Figure 9 Scatter Plot of Apple Centers

in the test set and the fact that this step does not necessitate calculating the actual number of apples, we opted for a model characterized by its speed and efficiency. After comparison, we found that the YOLOv5x model still possesses these advantages.

Initially, we used **LabelImg** to uniformly annotate all photos in *Attachment 2*. Due to the large number of these photo samples, we selectively established a dataset, adhering to the YOLOv5 recommended standard of 1,000 images per category. We then divided the dataset into training and testing sets in an 8:2 ratio, with training parameters consistent with those mentioned earlier, except the batch-size was 16 and the number of iterations was increased to 100. The training mainly involved extracting and differentiating features of five types of fruits in *Attachment 2*.

Subsequently, we tested the trained model with *Attachment 3*, and the results are shown in Figure 10.

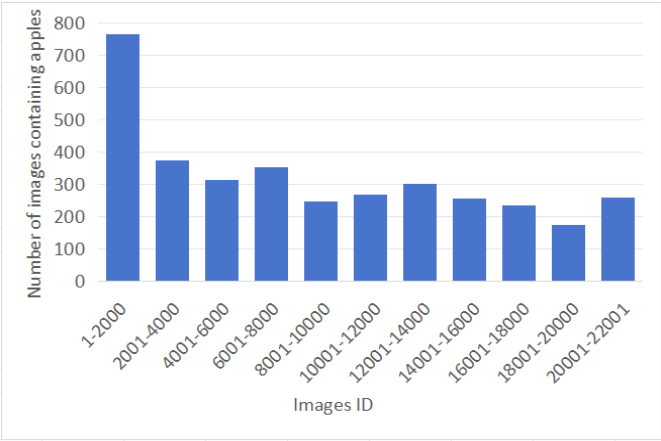


Figure 10 Distribution histogram of the ID Numbers of All Apple Images

IV. Model 2-Apple Maturity Detection Model based on Spectral Clustering

4.1 Description of the Model

Spectral clustering is a common clustering algorithm known for its strong adaptability to data distributions, excellent clustering results, and low computational requirements, making it a widely favored solution for clustering problems. It defines an optimization objective function for subgraph partitioning based on a data similarity matrix and has improved upon two partitioning methods: **RatioCut** and **NCut**.

The most commonly used spectral clustering algorithm consists of three steps: **a fully connected approach based on Gaussian kernel distances; the graph partitioning method NCut; and the clustering method K-Means**.

The fully connected approach requires that the weight values between all points are greater than zero, and the Gaussian kernel function RBF is commonly used to define edge weights. The formula for the similarity matrix and adjacency matrix under the fully connected method is as follows:

$$W_{ij} = S_{ij} = \exp\left(-\frac{\|x_i - x_j\|_2^2}{2\sigma^2}\right)$$

Here, W_{ij} represents the adjacency matrix, which is a matrix composed of the weight values w_{ij} between any two points; S_{ij} is the similarity matrix, and we usually obtain the adjacency matrix through an easily measurable similarity matrix. x_i and $x_j$represent any two points, and $\|x_i - x_j\|_2^2$ mainly indicates the Euclidean distance between the two points. σ is used to control the scope of the Gaussian kernel function, and its value should be set within a range that optimizes the model fitting effect.

After obtaining the adjacency matrix, we need to **partition the graph**. Here, we adopt the commonly used Normalized Cut method. To avoid the poor performance resulting from using the minimum cut method, the **NCut** method maximizes the weight of each node in the sub-graphs while minimizing the cut. The basic formula and the final derived formula are as follows:

$$\mathrm{NCut}(A, B) = \frac{\mathrm{cut}(A, B)}{\mathrm{vol}(A)} + \frac{\mathrm{cut}(A, B)}{\mathrm{vol}(B)}$$

$$\mathrm{argmintr}(F^T D^{-1/2} L D^{-1/2} F) \quad \text{s.t.} \quad F^T F = I$$

$$L = D - A$$

Here, L refers to the Laplacian matrix of the graph, D refers to the inverse matrix of the graph, and F is the matrix of eigenvectors of the graph, typically containing the first K eigenvectors obtained from the eigenvalue decomposition of the Laplacian matrix. The argmintr part $\text{argmintr}(F^T D^{-1/2} L D^{-1/2} F)$ of the formula represents the optimization objective of spectral clustering, while the $F^T F = I$ part is the constraint condition ensuring the orthogonality of the eigenvector matrix F, where I denotes the identity matrix.

4.2 Application of our model

In addressing **Question 3**, the spectral clustering algorithm offers the advantages of both efficiency and high precision. We treat all the apples in *Attachment 1* as distinct points distributed on the same undirected weighted graph, and establish a weighted connection between any two apples to test their similarity in various maturity features. The higher the similarity, the higher the weight coefficient. We then construct a similarity matrix to derive the adjacency matrix, which is used for subsequent graph partitioning and clustering tasks. The specific principle is illustrated in Figure 11.

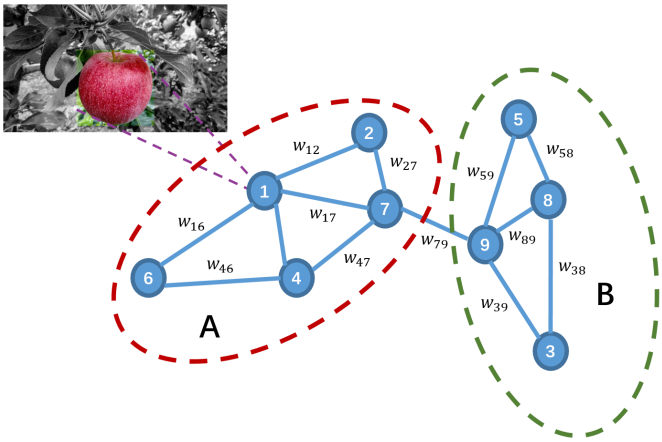


Figure 11 The Basic Principle of Spectral Clustering

Image data that influences the maturity of apples generally includes **blemishes, color, and size**. Upon observing the sample images in *Attachment 1*, we find that due to the low resolution, the model based on blemish detection is not effective in determining maturity, so this method is excluded. Moreover, determining apple maturity based on size is prone to interference from controlled variables such as variety. Considering these factors, we conclude that apple color can generally serve as a uniform standard for determining maturity.

We use the **HSV** color model to determine the color of the samples, which includes hue, saturation, and value. Among these, the value is greatly affected by the lighting of the sample images, resulting in less than ideal practical effectiveness. Therefore, we choose the frequency data of hue and saturation as indicators. In other words, **we focus more on whether the sample colors in the image are redder or greener, and more vivid or more dull**.

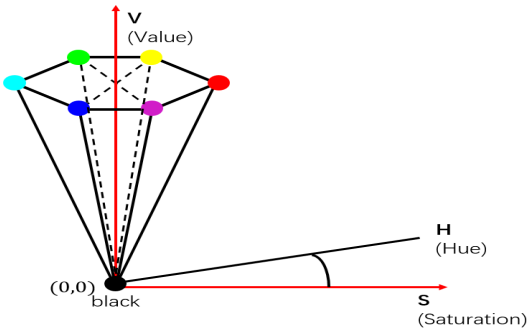


Figure 12 HSV Color Model

By maximizing the use of color information from all valid data points in the image, we have drawn a frequency distribution graph of the sample's HSV, some of which are as shown in Figure 13.

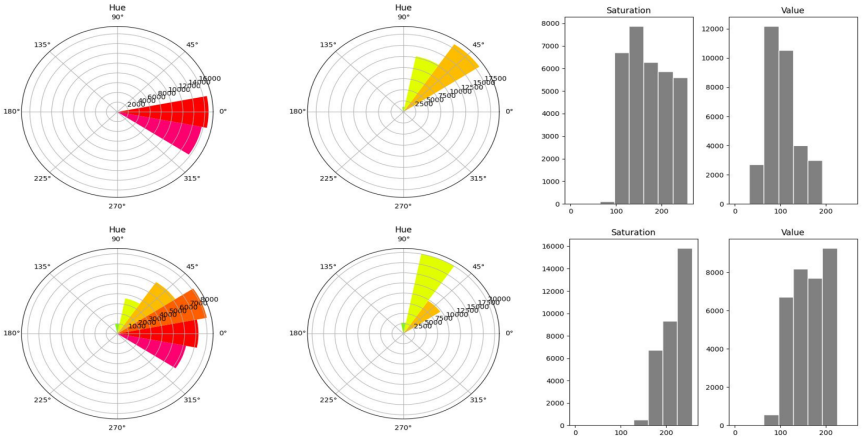


Figure 13 distribution graph of the sample's HSV

Given our outstanding performance in segmenting the apple dataset, we assumed the shape of the apple as **the largest ellipse tangent to the edges of the segmented image**, considering only the pixel color information within the ellipse. Additionally, we standardized the size of the images before adding them to the training set.

Subsequently, we transformed the frequency data of H and S into **feature vectors** related to apple maturity. We took the two-dimensional coordinate values of these

vectors as the points corresponding to the apples in an undirected weighted space, totaling 2568 points, and then performed graph partitioning and clustering analysis on these points. The reason for choosing the Spectral Clustering model for clustering is due to its sensitivity to **high-dimensional clustering distributions**.

4.3 Results

The result of the clustering is shown in **Figure 14**.

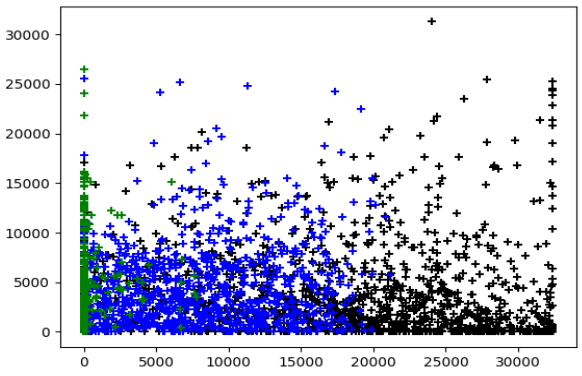


Figure 14 The Scatter Plot of the clustering

The histogram of the maturity distribution of all apples is as shown in **Figure 15**.

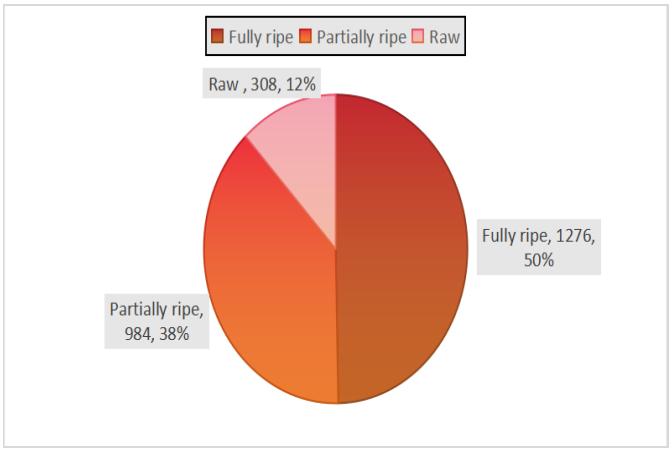


Figure 15 The Histogram of the Maturity Distribution of all Apples

V. Model 3-Multi-modal Monocular Depth Estimation & Quality Measurement Model

5.1 Apple Volume Estimation based on ZoeDepth

5.1.1 *Depth Estimation Theory*

Depth estimation refers to obtaining the distance information of each point in a scene from the camera in an image. Monocular depth estimation is one such method, primarily estimating the distance of each pixel in an image relative to the shooting source from a single picture. This method assumes that, similar to how the human eye can extract spatial distance information from captured images, monocular depth estimation can achieve the same.

Research on monocular depth estimation is mainly divided into two schools: **absolute depth estimation** and **relative depth estimation**. Absolute depth estimation refers to estimating the depth in absolute physical units. This method is limited to a single camera model and cannot achieve mixed training due to scale ambiguity. Relative depth estimation also has the drawback of being unable to recover the real-world scale.

Scale ambiguity often refers to the misjudgment of image depth that occurs under different camera parameters, highlighting the importance of camera internal parameters for the estimation accuracy of a single image. However, factors like sensor size and pixel dimensions do not affect depth estimation. As shown in Figure 16, despite the different image resolutions, the distance of objects from the camera remains consistent.

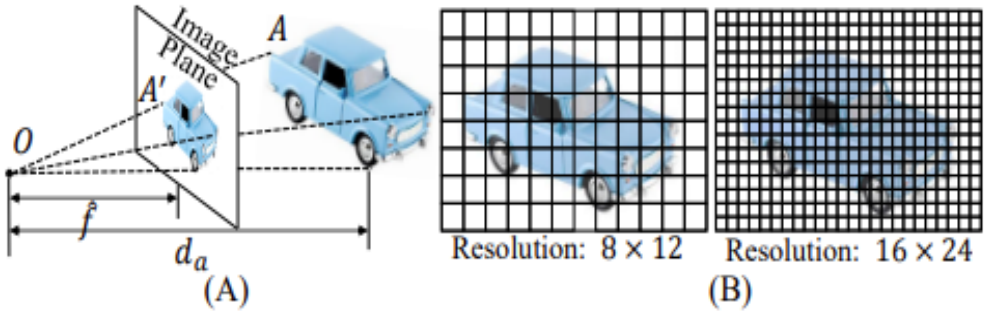


Figure 16 Depth Estimation Theory [3]

5.1.2 *Description of ZoeDepth*

Given the drawbacks of the two methods mentioned above, the authors of the **ZoeDepth** paper proposed a hybrid approach that combines the advantages of good generalization performance and maintaining metric scale.

This model primarily operates through **a two-stage framework**. In the first stage, a general encoder-decoder architecture is used for relative depth estimation pre-training. In the second stage, a metric bins module for absolute depth estimation is added, followed by fine-tuning using an absolute depth dataset.

This module ultimately predicts the center values of N_total bins for each pixel depth interval. A **'bin'** refers to the intervals classified by discretizing the continuous depth range. The final depth of each pixel is calculated using the following formula:

$$d(i) = \sum_{k=1}^{N_{\text{total}}} p_i(k) c_i(k)$$

Here, $d(i)$ refers to the final depth of the i pixel, $c_i(k)$ is the center value of the bin, and $p_i(k)$ is the probability of that bin center value.

5.2 Difficulties & Solutions

We use a combined model based on two-dimensional area and depth estimation of the actual area to estimate the weight of the apple. We selected the PFuji-size dataset from the Zenodo website for pre-training. This dataset contains a total of 615 Fuji apples scanned in-field conditions and an additional 25 apples scanned in laboratory conditions. We chose Fuji apples because they are nearly spherical in shape and uniformly red throughout, which is advantageous for subsequent processing.

Before training, we pre-processed the samples for 3D point cloud generation, with the results shown in Figure 17.

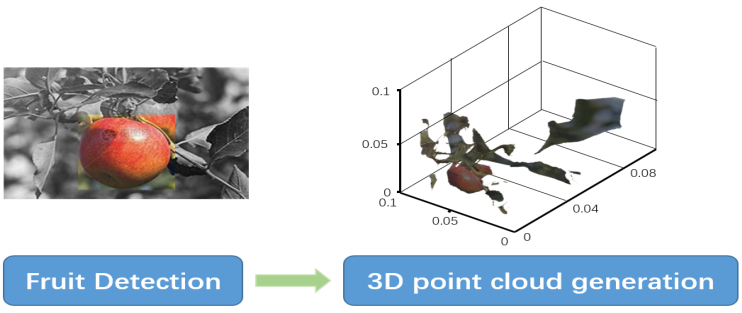


Figure 17 Data Preprocessing

In estimating the actual size of apples, we encountered certain limitations in the sample images provided in the attachment, which created some difficulties for our estimation work. The main challenges include:

• Unknown camera parameters: The various photos in the problem statement do not reveal the external parameters (distance and angle of the camera relative to the object) or internal parameters (lens focal length) of the camera.

• Absence of reference objects with known actual sizes in the images.

• All images have been uniformly processed to the same size, so the actual pixel size is unknown.

• The images come from various sources, with differing actual shooting locations, lighting conditions, etc.

Given these numerous limitations, accurately estimating the actual size of objects is quite challenging. We combined the current forefront research directions in the field of Computer Vision and attempted to measure the volume of apples starting from the algorithm of monocular absolute depth estimation.

In terms of hyperparameter optimization, after the dataset processing and training, we finally adjusted the learning rate to 0.0002512 and set the weight decay exponent to 0.01. The model exhibited superior performance upon evaluation.

Subsequently, we tested the samples from Attachment 1 in the dataset, ultimately outputting depth images for each sample and a data file corresponding to all apples in the samples. This data file includes the distance of each apple, the width and height of the target bounding box, the fruit shape index, and the average diameter of the apples. Some of the data files are displayed in the following table, and all data files will be submitted in the appendix.

Apple ID	Image ID	depth	width	height	aspect_ratio
0	1	1.9235	36	29	0.8056
1	10	1.4657	55	55	1
2	10	8.8861	7	8	1.1429
3	10	5.0029	9	9	1
4	10	9.5632	11	8	0.7273

Table 4 The Depth Estimation of All Apples

It's worth noting that the Image IDs in this table are not arranged in sequential order. And

$$aspect_ratio = \frac{height}{width}$$

The image results of depth estimation are shown in **Figure 18**.

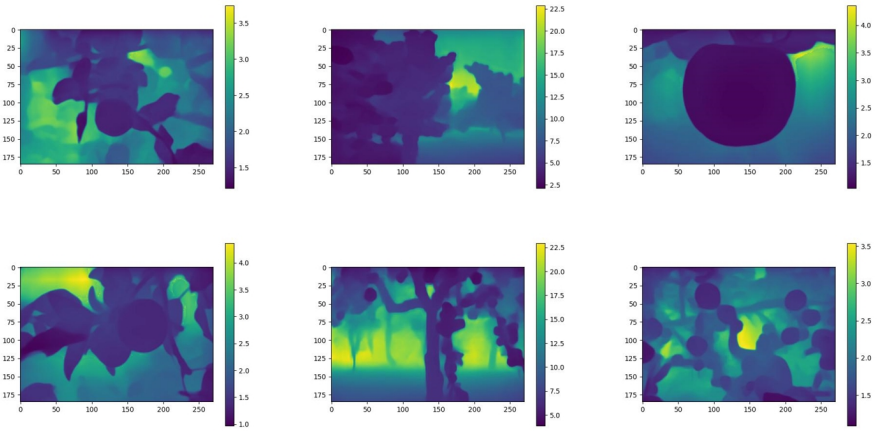


Figure 18 depth estimation images

Then we proceeded to obtain the depth values of the target samples. We first took **the central pixel of the bounding box** for the distance corresponding to the center point of the apple (i.e., the metric depth from the camera lens). For non-integer central values, we used an integer floor function. Then, we located the corresponding distance value in the predicted depth array file *np.array* outputted by **ZoeDepth**. We specifically chose Model **ZoeDepth-M12-NK**, as it demonstrated the best performance.

Ultimately, we obtained depth estimations for all the apples in *Attachment 1*. We drew a **depth distribution box plot** for a more intuitive display of the depth distribution of all apples, as shown in Figure 19.

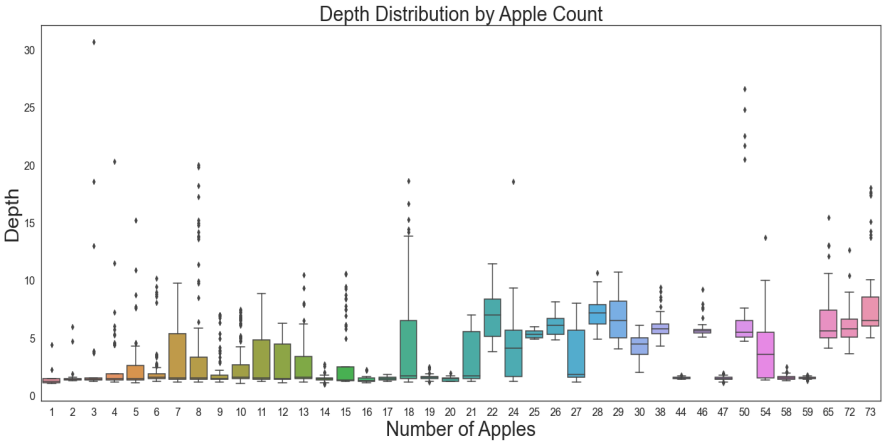


Figure 19 depth distribution box plot

From analyzing the image features, we can observe that generally, images with fewer apples have a smaller average depth. This might be because most of the close-up images in the sample primarily feature a single apple, while the long shots are dominated by apple trees, which obviously contain a larger number of apples. Additionally, due to the varying positions of apples within a single image, there are many long-tail outliers in the box plot.

Further analysis reveals that the greater the depth, the smaller the average pixel diameter of the apples in the image. This characteristic is displayed in Figure 20.

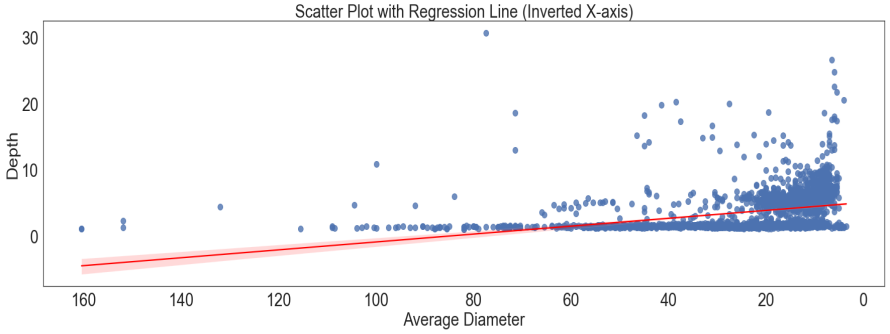


Figure 20 Scatter Plot with Regression Line

5.3 Results

First, we calculate the two-dimensional area using the data obtained from Depth Estimation. The calculation formula is as follows.

$$S_{td} = \pi \frac{R^2}{2}$$

Where S_{td} means the two-dimensional area of apples.

Then we present the descriptive statistics of the calculated two-dimensional area as shown in the table.

Subsequently, we calculated the actual area using the **Shape Index** (SI), which refers to the ratio of the fruit's vertical diameter to its horizontal diameter. Due to the large error in volume calculation using two-dimensional area, we opted to enhance the prediction accuracy by incorporating a method to estimate the actual volume of the fruit.

We assume the apple's cross-section as the circumscribed circle or ellipse of the bounding box, with the height in the image pixels representing the vertical diameter, and the width representing the horizontal diameter.

$$height = y_{max} - y_{min}$$

Index	Value
count	2568
mean	711.880552
std	1655.982606
min	9.424778
max	46759.465056

Table 5 Descriptive Statistic of Pixel Area

$$width = x_{max} - x_{min}$$

From the definition, we know that the closer the SI is to 1, the closer the apple is to a perfect circle. Observing the SI distribution chart (**Figure 21**), we find that the average SI is 0.955, with most SI values close to 1.

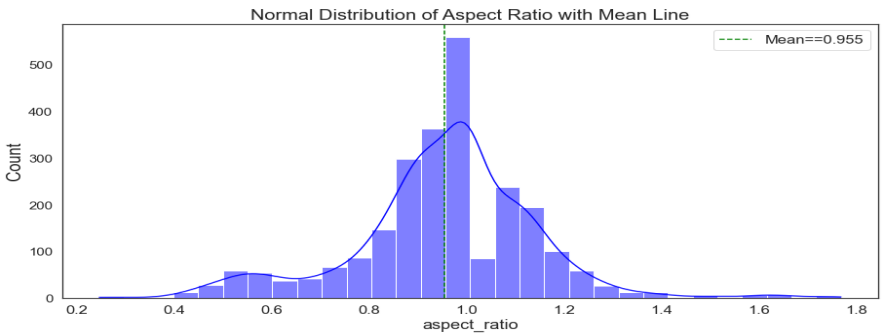


Figure 21 The Histogram of SI distribution

Since the bounding box might be incompletely marked due to leaf occlusion during labeling, we specially processed apples with abnormal SI (less than 0.5 or more than 1.5), taking the $max(height, width)$ and defaulting to a circular shape.

We calculated the apple's area using geometric formulas and estimated its volume based on the area and the scaling ratio of the image. The formulas for calculating the area and volume are as follows:

$$V_{Apple} = aR_r^3 + bR_r^2 \begin{cases} R_r = \frac{height+width}{2}, & \text{if SI is in [0.5-1.5]} \\ R_r = \max(height, width), & \text{else} \end{cases}$$

$$R_r = \frac{R}{MetrixDepth} * ScalingRatio$$

Where V_{Apple} represents the volume of apples. a/b represent model coefficients. Through reviewing relevant literature [4] and repeated training simulations of the model, we determined that the value of a is 0.0002, and the value of b is 0.0249. R_r represents the real R.

The R_r formula is derived from the focal length formula (based on the principle of pinhole imaging). Since the focal length formula is only applicable to the quasi-focal plane, if the distance increases, the relative focal length also increases, leading to an increased scaling ratio within the same plane.

Regarding the setting of proportions, we classify the images into **close-up (d<5), medium-shot (d=5-20), and long-shot (d>20)** based on the average depth of the apple centers in the image. We set a specific scaling ratio for each scene and make slight adjustments to this ratio based on the specific conditions of the image.

Subsequently, we use the mass formula to convert volume into density. We categorize apples into three different densities based on their maturity level as determined in Q3: mature apples have a density of 0.88, semi-mature apples 0.84, and immature apples 0.8.

The histogram of Masses of all apples we drew are shown as **Figure 22**.

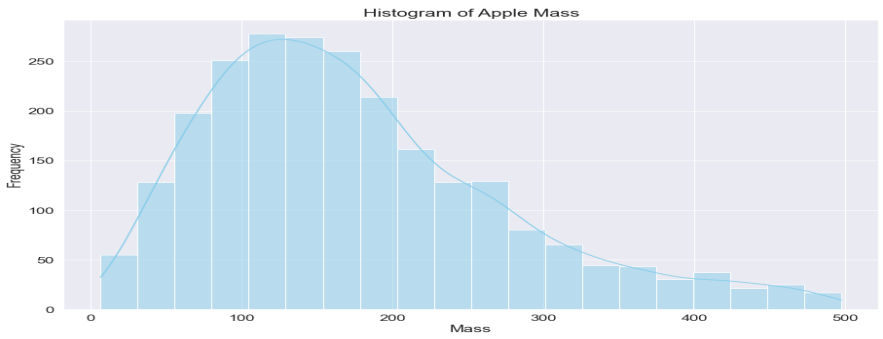


Figure 22 The Histogram of Masses of all Apples

VI. Interesting Findings

During the process of solving the apple recognition problem, we encountered some **interesting phenomena**. By conducting large-scale training and prediction with the sample data provided in Attachment, we derived a series of related statistical conclusions. These conclusions show unexpected correlations at certain levels, and some of these findings may lead us in different directions in future research.

In estimating the depth of apples in sample images, we found that **images with**

fewer apples tend to have a smaller average depth. This could be because most close-up images in the sample primarily feature a single apple, whereas long shots are dominated by apple trees, which contain a significantly larger number of apples. The difference in shooting subjects influences the photographer's choice of close-up or long shots. This concept at the photography level is reflected in our depth measurement model, corroborating the model's accuracy from a practical perspective.

VII. Model Assessment & Further Discussion

For the evaluation of **Model 1**, we used common metrics such as precision, recall, and F1 score. A detailed evaluation is provided in section **3.4.3** and is not repeated here.

In **Model 2**, we focused on the H and S values of the HSV color, selecting the clustering information of high-saturation pixels under the red hue. As shown in the figure, the model's clustering boundary clearly reflects that the characteristics of apples with a red hue are significantly different from other apples, which aligns with our common sense. However, limited by the resolution of the original data images, we were unable to extract and analyze detailed features and could not completely eliminate the loss due to leaf occlusion. In global research, a large number of image restoration algorithms based on baseline have gradually matured. Perhaps we can start with image restoration algorithms to extract detailed features of the obscured parts of apples, and thus make a more detailed assessment of apple maturity.

In **Model 3**, the image depth assessment technology helped us to restore the size of apples, which was crucial for estimating their weight. By calculating density in stages, effectively using known color information and maturity, the systematic error in our apple mass estimation will be further reduced.

VIII. References

[1] Chen Qing, Yin Chengkai, et al. *Current status and future development of the key technologies for apple picking robots* [J], Transactions of the Chinese Society of Agricultural Engineering, 2023, 39(04), 1-15.

[2] Wang peng. *Research on Apple Fruit Recognition Based on Improved YOLO Algorithm* [D]. Jilin Agricultural University, 2023.

[3] Wei Yin, Chi Zhang, et al. *Metric3D: Towards Zero-shot Metric 3D Prediction from A Single Image*, https://arxiv.org/abs/2302.12288v1, 2023/11/26.

[4] Liu Zhongchao, Fan Lingyan, et al. *Study on apple weight detection based on machine vision* [J]. Jiangsu Agricultural Sciences, 2021, 49(21), 201-205.

IX. Appendix

Listing 1: Target Box Positioning(Resource from YOLO)

```
# Filter the rows where class is 0 (which represents apples)
apple_df = df[df['Class'] == 0]

# Count the number of apples in each image
apple_counts = apple_df['Image Path'].value_counts()

# Define the bin size for the histogram
bin_size = 1 # This value determines the range each bin will cover

# Create bins starting from the minimum number of apples to the maximum
# and incrementing by the bin size

plt.figure(figsize=(10, 6))
apple_counts.hist(edgecolor='black', bins=[0, 10, 20, 30, 40, 50, 60,
    70])
plt.xlabel('Number of Apples', fontsize=20)
plt.ylabel('Frequency', fontsize=20)
plt.xticks(fontsize=16)
plt.yticks(fontsize=16)
plt.savefig('omg.png',dpi=500)
plt.show()

image_width = 270
image_height = 185

df['Y_min_transformed'] = image_height - df['Y_max']
df['Y_max_transformed'] = image_height - df['Y_min']

df['center_x'] = (df['X_min'] + df['X_max']) / 2
df['center_y'] = (df['Y_min_transformed'] + df['Y_max_transformed']) / 2

df[['center_x', 'center_y']].head()
```